

Project

Plant Care Tracker

**Optimization-Based System
for Indoor Plants**



Introduction of Project Idea

Today, we are presenting our project Plant Care Tracker, which is an optimization-based system designed to help users manage indoor plants more efficiently.

Indoor plants improve the quality of living spaces, but managing them can be challenging, especially when users have multiple plants with different needs.



Problem Overview

Many users struggle with organizing plant care due to several issues, such as:

- Forgetting watering schedules
- Placing plants in unsuitable locations
- Managing multiple plants at the same time

Without a structured system, plant care becomes inefficient and may lead to unhealthy plants.

Problem Description

The main problem is the lack of an organized system that helps users track plant requirements and schedule maintenance tasks effectively.

Managing plant care involves multiple factors such as:

- Sunlight exposure
- Watering frequency
- Available space
- User time

When the number of plants increases, managing these factors manually becomes difficult.



Project Objectives

The objective of this project is to develop a system that:

- Tracks plant information and care requirements
- Recommends suitable locations based on light conditions
- Schedules plant care tasks efficiently
- Reduces user effort in managing plants

Constraints & Assumptions

The system considers several constraints, including:

- Limited indoor space
- Different lighting conditions
- Limited time for plant care

These constraints make the problem suitable for an Operations Research solution.



Why Operations Research?

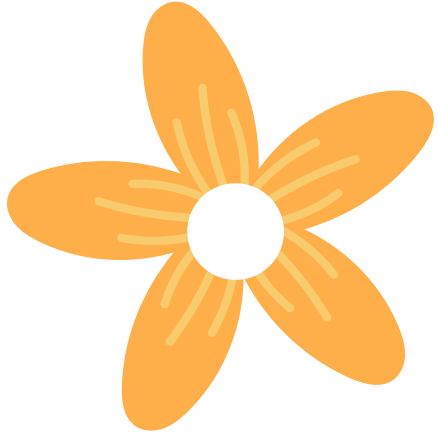
This problem is suitable for Operations Research because it involves decision-making under constraints.

By applying optimization techniques, the system can:

- Improve plant placement decisions
- Organize care schedules
- Provide optimal solutions instead of manual guessing



Methodology



Operations Research Approach

- The system applies an Operations Research (OR) model to optimize plant care decisions.
- A real dataset is used to define model parameters and ensure realistic results.

Model Type: Binary Linear Programming (BLP)

- The model is based on binary decision-making, where:
 - Plant Placement → assigned (1) or not assigned (0)
 - Care Scheduling → scheduled (1) or not scheduled (0)

Key Idea

- The goal is to optimize plant management by:
 - Selecting the most suitable location for each plant
 - Organizing care tasks efficiently

Mathematical Model

Decision Variables

x_{ij}

→ = 1 if plant i is assigned to location j

→ = 0 otherwise

y_{it}

→ = 1 if plant i is scheduled for care on day t

→ = 0 otherwise

Objective Function

$$Z = \alpha \sum S_{ij}x_{ij} + \beta \sum P_i y_{it}$$

Variables Description

Z : Total system performance

S_{ij} : Suitability score (plant–location)

P_i : Priority of plant i

α : Importance of placement

β : Importance of scheduling



Mathematics Model – Constraints

1. Plant Assignment

$$\sum x_{ij} = 1 \quad \forall i$$

Each plant is assigned to exactly one location

2. Location Capacity

$$\sum x_{ij} \leq C_j \quad \forall j$$

Each location has limited capacity

3. Care Frequency

$$\sum y_{it} = W_i \quad \forall i$$

Each plant must receive required care sessions

4. Daily Workload

$$\sum y_{it} \leq L_t \quad \forall t$$

Daily tasks must not exceed limit

5. Binary Constraints

$$x_{ij}, y_{it} \in \{0, 1\}$$

Decisions are binary (yes/no)

Parameters Description

C_j : Capacity of location j

W_i : Required care sessions

L_t : Maximum tasks per day



IMPLEMENTATION OF OPTIMIZATION MODEL IN EXCEL

Points:

- Model built using OpenSolver
- Objective: Maximize total objective value
- Constraints: Capacity, demand, and daily limits
- Output: Optimal assignment and scheduling

The screenshot shows the 'OpenSolver - Model' dialog box. At the top, there is a section 'What is AutoModel?' with a description and an 'AutoModel' button. Below this, the 'Objective' is set to '\$C\$28' with a dropdown arrow, and the 'maximise' radio button is selected. The 'Variable' is set to '\$B\$2:\$C\$4'. The 'Constraints' section lists several constraints: '\$E\$11:\$E\$13 <= \$F\$11:\$F\$13', '\$D\$2:\$D\$4 = 1', '\$B\$2:\$C\$4 bin', '\$B\$16:\$B\$18 <= \$C\$16:\$C\$18', and '\$B\$7:\$B\$8 <= \$C\$7:\$C\$8'. There are input fields for adding new constraints and buttons for 'Add constraint', 'Cancel', and 'Delete selected constraint'. The 'Sensitivity' section has checkboxes for 'List sensitivity analysis on the same sheet with top left cell:' and 'Output sensitivity analysis:', with the latter set to 'updating any previous output sheet'. The 'Solver Engine' is set to 'CBC'. At the bottom, there are checkboxes for 'Show model after saving' and buttons for 'Clear Model', 'Options...', 'Save Model', and 'Cancel'.



IMPLEMENTATION OF OPTIMIZATION MODEL IN EXCEL

Points:

- Suitability matrix between plants and locations
- Binary decision variables (0 or 1 assignment)
- Capacity constraints for each location
- Demand (required vs scheduled) satisfaction
- Daily task limits (max tasks per day)
- Objective function results (Total = 35)

[illegible]



Experiments

Model tested under different scenarios
Evaluate model performance under changes

Scenarios:

- Scenario 1: Base case (original data)
- Scenario 2: Increased care frequency ($W_i + 1$)
- Scenario 3: Reduced location capacity (C_j)

Results

- Scenario 1:

Optimal solution ($Z = 35$)

- Scenario 2:

Feasible solution

High objective value achieved

- Scenario 3:

Limited feasible assignments due to capacity restrictions



Analysis & Insights

- Model is flexible in scheduling
- Can handle increased care demand
- Sensitive to space limitations
- Capacity is the key constraint
- Higher objective value indicates improved system performance



Optimal Solution

- The solution is optimal because all constraints are satisfied
- No constraint is violated in the final solution
- Excel Solver evaluated all possible combinations
- No better solution can be found without violating constraints





Objective Function



- Optimal value: $Z = 35$
- Represents total system performance
- Combines:
 - Plant placement suitability
 - Care scheduling efficiency
- Higher value = better decision quality

Conclusion

This project applied a Binary Linear Programming model to optimize plant care management. The system achieved an optimal and feasible solution while satisfying all constraints. The results highlight the effectiveness of optimization techniques in improving decision-making.





Thank You

